Figure 3: The Huxleyan Reducing Valve & K.MOSH Conscious Observer Architecture

Eliminative Sensory Compression, Unified Intra-Psychic Monologue (A' ⇔ B'), 3-Tier Epistemic Lifecycle, and Top-Down Causal Propagation

1. Subconscious Sensory-Motor Substrate & Huxleyan Reducing Valve (HuxleyReducingValve)

10^900 RAW STATES → PHENOMENOLOGICAL STREAM

Cross-Pair Carrier & Wall Topology

- Extracts canonical background carrier **c_bg**
- Identifies full-span divider walls **C_wall**
- Detects 4×4 hollow/solid & 5-wide/tall frames
- Eliminates 99.9% of ungrounded states in <2 ms

S10 Ontological Color-Role Extraction

- 8-connected components outside wall dividers
- Binds monochromatic anchor **marker_col**
- Binds multi-color sprite **body_col**
- Invariant to arbitrary **π ∈ S_10** palette permutations

D4 Orientation & Flow Vector Cues

- Perimeter flow vectors `edge_flow[top,bot,L,R]`
- Aspect-ratio parity (`in_dims → out_dims`)
- Synthesizes priority orbit **π_D4 ∈ D_4***
- Emits reduced tuple **S_phenom = R_valve(D_train)**

Compressed Phenomenological Stream S_phenom

Recalled LTP Archetype Priors

2. Unified Conscious Observer (KMoshConsciousObserver)

MISTRAL-NEMO-8B (LOCAL GGUF)

STREAM A' (INTUITION) First-Person Experiential Gestalt

"I see multi-colored sprites and isolated markers on background c_bg separated by walls, or a stream originating from the bottom/left edge with scattered obstacles in its path."

STREAM B' (SELF-AUDIT) Intra-Psychic Premise Critique ("One Being")

"Wait — my habitual subconscious reflex assumed downward-only gravity or hardcoded palette colors 8 and 9. Under D4 rotation or S10 permutation, I must inhibit that reflex, re-center my reference frame via g ∈ D4, and bind marker_col and body_col by topology."

Consolidated Belief & Downward Causal Directive (D_↓)

Emits narrative + **`d4_priority_order`** + **`s10_role_bindings`** + **`inhibited_assumptions`**

Zero bureaucratic tool RPCs; directly reconfigures subconscious perceptual lenses

3. Three-Tier Epistemic Memory Lifecycle

LTP & FORGETTING

Tier 1: Ephemeral Working Thoughts (Miller 7 ± 2 Buffer)

- Bounded priority queue **|T_work| ≤ 7** (`EphemeralIdea`)
- Ranked by residual energy E(I); contradicted ideas evicted first

Tier 2: Falsification-Verified Working Beliefs

- Promoted only when top-down closure achieves **Δ_train = 0**
- Binds verified `(archetype, winning_op, d4_frame, role_map)`

Tier 3: Kandel LTP Episodic Memory & Graceful Forgetting

- Synaptic boost on hit: **w_k ← min(5.0, w_k + 1.15)** (w_0 = 1.5)
- Exponential decay on disuse: **w_k ← 0.88 · w_k**
- Pruned when **w_k < θ_forget = 0.25** (prevents catastrophic interference)

Downward Causal Directive D_↓ (Frame & Role Re-Centering)

Δ_train = 0 Verification → LTP Consolidation

4. Top-Down Causal Affordance Propagation (propagate_top_down_affordances)

$f_{conj}(X) = g^{-1}(f_\rho(g(X)))$ WITH Δ_train = 0

Dihedral Reference Frame Conjugation

- Conjugates training pairs **(g(X_m), g(Y_m))**
- Re-synthesizes innate tools in frame **g ∈ D_4**
- Opens orientation-gated subgrid/stream lenses
- Pulls prediction back via inverse **$g^{-1}(f(g(X)))$**

S10 Role-Equivariant Operator Binding

- Replaces hardcoded color literals (0, 8, 9) with inferred `bg_color, marker_col, body_col`
- Orders polyomino packing pieces by geometry
- Evaluates across all 32 verified operator families

Falsification Gate & Skill Promotion

- Admits closure `kmosh_topdown_{g}__{fn}` iff **$\sum \|g^{-1}(f(g(X_m))) \ominus Y_m\|_0 = 0$**
- Promotes closure to `newly_learned_skills`
- Rescues 100% (10/10) of D4×S9 conjugated OOS tasks

FALSIFICATION-CERTIFIED TOP-DOWN PREDICTION & LTP TRACE

100.0% Tier-1 Synthetic OOS | 100.0% Tier-2 D4×S9 Conjugated ARC-2 OOS | Air-Gapped Container Verified